

The Program in Action

Figure 35-1 shows what the output will look like when you run *hexgrid.py*.

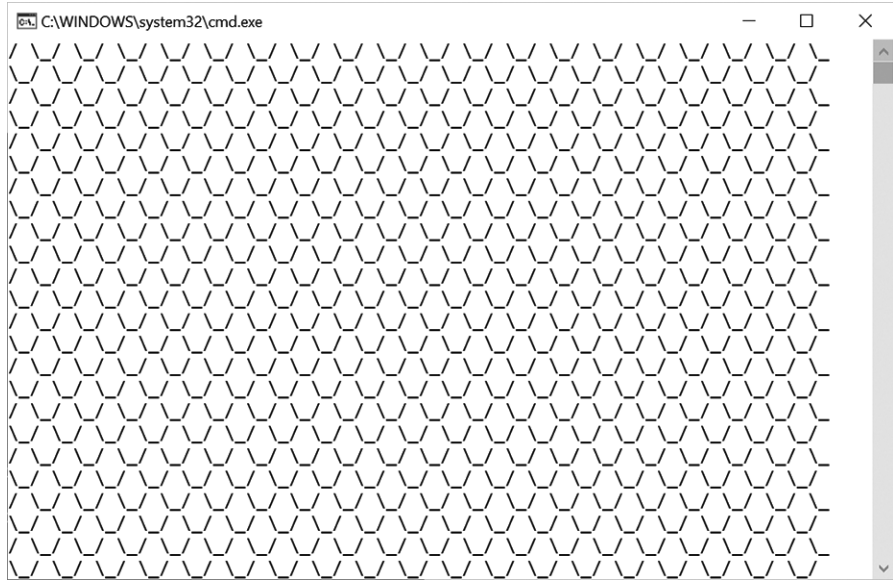


Figure 35-1: Output showing a tessellated image of a hexagonal grid

How It Works

The power behind programming is that it can make a computer carry out repetitive instructions quickly and without mistakes. This is how a dozen lines of code can create hundreds, thousands, or millions of hexagons on the screen.

In the Command Prompt or Terminal window, you can redirect a program's output from the screen to a text file. On Windows, run `py hexgrid.py > hextiles.txt` to create a text file that contains the hexagons. On Linux and macOS, run `python3 hexgrid.py > hextiles.txt`. Without the size of the screen as a limit, you can increase the `X_REPEAT` and `Y_REPEAT` constants and save the contents to a file. From there, it's easy to print the file on paper, send it in an email, or post it to social media. This applies to any computer-generated artwork you create.

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1. `"""Hex Grid, by Al Sweigart al@inventwithpython.com`
 2. `Displays a simple tessellation of a hexagon grid.`
 3. `View this code at https://nostarch.com/big-book-small-python-projects`
 4. `Tags: tiny, beginner, artistic"""`
 - 5.
 6. `# Set up the constants:`
 7. `# (!) Try changing these values to other numbers:`
 8. `X_REPEAT = 19 # How many times to tessellate horizontally.`
 9. `Y_REPEAT = 12 # How many times to tessellate vertically.`
 - 10.